

1. Let $\{f_n\}_{n \in \mathbb{N}}, \{g_n\}_{n \in \mathbb{N}}, \{h_n\}_{n \in \mathbb{N}}$ be sequences of real valued Lebesgue integrable functions which converge almost everywhere to f, g, h respectively. Additionally suppose that $f_n \leq g_n \leq h_n$ for all n and that $\lim \int_{\mathbb{R}} f_n d\mu = \int_{\mathbb{R}} f d\mu, \lim \int_{\mathbb{R}} h_n d\mu = \int_{\mathbb{R}} h d\mu$. We note that $0 \leq g_n - f_n$ and $0 \leq h_n - g_n$. Therefore we have that

$$\begin{aligned} \int g_n d\mu - \int f_n d\mu &= \int (g_n - f_n) d\mu = \int \lim (g_n - f_n) d\mu \\ &\leq \liminf \int (g_n - f_n) d\mu = \liminf \int g_n d\mu - \int f_n d\mu \\ \int h_n d\mu - \int g_n d\mu &= \int (h_n - g_n) d\mu = \int \lim (h_n - g_n) d\mu \\ &\leq \liminf \int (h_n - g_n) d\mu = \int h_n d\mu - \limsup \int g_n d\mu \end{aligned}$$

Therefore we have that $\int g \leq \liminf \int g_n$ and $\int g \geq \limsup \int g_n$. Thus $\int g d\mu = \lim \int g_n d\mu$

2. Let $\{f_n\}_{n \in \mathbb{N}} \subset L^+(X, \mathcal{M}, \mu)$ such that $f_n \rightarrow f$ pointwise and such that $\lim \int_X f_n d\mu = \int_X f d\mu < \infty$. Let $E \in \mathcal{M}$. Note that $1_E |f - f_n| \leq |f - f_n|$ for all $n \in \mathbb{N}$. Therefore $\lim_{n \rightarrow \infty} \int_E |f - f_n| d\mu \leq \lim_{n \rightarrow \infty} \int_X |f - f_n| d\mu = 0$. Therefore $\lim_{n \rightarrow \infty} \int_E |f - f_n| d\mu = 0$ which implies that $\lim_{n \rightarrow \infty} \int_E f_n d\mu = \int_E f d\mu$. Note that if $\int f d\mu = \infty$ then everything collapses since f is now not in L^1 or almost everywhere equivalent to something in L^1 , which is what the entire proof is based upon.

3. Let μ be the Lebesgue measure on $\mathbb{R}$, $f \in L^1(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ and for $n \in \mathbb{N}$ define $f_n(x) = f(x) \sin^n(x)$. We want to show that $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n d\mu = 0$. We claim that $f'(x) =$

$$\lim_{n \rightarrow \infty} |f_n(x)| = \begin{cases} |f(x)| & x = \pi(n + \frac{1}{2}), n \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}. \text{ Note if } x \neq \pi(n + \frac{1}{2}), n \in \mathbb{Z}, \text{ we have}$$

that $|\sin(x)| < 1$. Therefore $\lim |\sin^n(x)| = 0$. Otherwise, if $x = \pi(n + \frac{1}{2})$ then we have that $|\sin(x)| = 1$. Therefore $\lim |f(x) \sin^n(x)| = |f(x)|$. Additionally since for any $n \in \mathbb{N}$, $|f(x) \sin^n(x)| \leq |f(x)|$, and $|f(x)| \in L^+(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ then by the LDCT we have that $\int_{\mathbb{R}} f'(x) d\mu = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} |f_n(x)| d\mu$. Note that since f' is non-zero on a countable set, then necessarily $\int_{\mathbb{R}} f'(x) d\mu = 0$. Thus $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n(x) d\mu = 0$.

4. Let $\{f_n\}$ be a sequence of integrable functions from $[0, 1]$ to $[0, \infty]$ with the property that $\int_{[0, 1]} f_n d\mu = 1, \int_{(\frac{1}{n}, 1]} f_n d\mu \leq \frac{1}{n}$. We claim that it is impossible for $g(x) = \sup_{n \in \mathbb{N}} f_n(x)$ to be integrable. Assume not. Note by the inequality given to us we have that $\int_{[0, 1/n]} f_n d\mu \geq 1 - \frac{1}{n}$. Therefore we have that $\lim \int_{[0, 1/n]} f_n d\mu = 1$. Additionally, note that $f_n 1_{[0, \frac{1}{n}]}$ are integrable functions. Furthermore, $\lim f_n 1_{[0, \frac{1}{n}]} = 0$ almost everywhere since for every $x \in (0, 1]$ there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x$. Additionally, since g is integrable and we know that $|f_n(x) 1_{[0, 1/n]}(x)| \leq g(x)$. Therefore by LDCT $\lim \int_{[0, 1]} f_n 1_{[0, 1/n]} d\mu = 0$. This is a contradiction since it was previously established that the limit was 1.

5. Let $f_n(x) = \frac{1}{1+e^{nx}}$. Note that on $[0, 1]$ that $\lim f_n = 1_{[0,1]}$ almost everywhere. Additionally $\lim \delta(x)f_n(x) = \delta(x)1_{[0,1]}(x)$ holds for all $x \neq 0$. Furthermore $|\delta(x)f_n(x)| \leq |\delta(x)1_{[0,1]}|$ as $f_n(x) \leq 1$, and the contribution of $\delta(x)$ doesn't matter. Therefore $0.5 = \lim \int_{[0,1]} |f_n(x)\delta(x)|d\mu = \int_{[0,1]} \delta d\mu = 1$, which is a contradiction.
6. Let $\{f_n\}$ be a sequence in $L^1(X, \mathcal{M}, \mu)$ such that $f \rightarrow f_n$ pointwise and there exists some function $g \in L^1(X, \mathcal{M}, \mu)$ such that $|f_n| \leq g$ for all $n \in \mathbb{N}$, and $\epsilon > 0$. Note that our assumptions give us by the LDCT that $\lim \|f - f_n\|_1 = 0$. Since the sequence converges in L^1 then there necessarily exists a sequence n_k such that for all $n, m \geq n_k$ then $\|f_n - f_m\|_1 < 2^{-k}$. Additionally let $D_k = \{x \in X : |f_{n_k} - f_{n_{k+1}}| > 2^{-k/2}\}$. Observe that $\mu(D_k)2^{-k/2} < \|f_{n_k} - f_{n_{k+1}}\|_1 < 2^{-k}$. Therefore $\mu(D_k) < 2^{-k/2}$. Furthermore if we take the measure of $\bigcup_{k=N}^{\infty} D_k$ then we have that $\mu(\bigcup_{k=N}^{\infty} D_k) \leq \sum_{k=N}^{\infty} \mu(D_k) < \sum_{k=N}^{\infty} 2^{-k/2} = \frac{2^{1-\frac{N}{2}}}{2-\sqrt{2}}$. Note that we can find a sufficiently large N_ϵ such that $\mu(\bigcup_{k=N}^{\infty} D_k) < \epsilon$. Thus $E_\epsilon = \bigcup_{k=N}^{\infty} D_k$.
Furthermore, to show that there is uniform continuity on the complement, we have that if $x \in (\bigcup_{k=N}^{\infty} D_k)^c$ then we have for every $n \geq N$ that $|f_n(x) - f(x)| = |f_n(x) - f_{k_N}(x) + \sum_{l=N+1}^{\infty} |f_{k_l}(x) - f_{k_{l+1}}(x)| < 2^{-N} + \frac{2^{1-\frac{N+1}{2}}}{2-\sqrt{2}}$